

HYDRA Documentation

HYDRA MW4.0pe Installing AIP 8.2 on a Terminal PC

Version 1.0.23049

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1 Purpose

This manual describes the necessary tasks to install the HYDRA Terminal software AIP 8.2 (Acquisition & Information Panel) on a standard PC hardware.

AIP 8.2 (AIP2) is especially designed to run on industrial PCs with a touch screen monitor and without a standard keyboard connected.

If the standard PC hardware and it's configuration does not meet the specifications described in this manual, certain functionalities of the HYDRA Terminal software might not work as advertised.

This manual covers the initial installation of the AIP 8.2 software on a standard PC hardware.

HYDRA MW4.0pe does not support older HYDRA terminal software than AIP 8.2.

If you have still older versions running in your environment, e.g. CTWIN or AIP 8.1, you need upgrade them first.

The manual **AIP2_Rollout_EN.pdf** describes the procedures to change the software on a PC still using CTWIN or AIP 8.1 (AIP) to AIP 8.2 (AIP2).

Please use the installation manual **InstallationGuide_MW40_AIP2_Citrix.pdf** if you plan to install the AIP 8.2 software on a Windows terminal server like Citrix XenAppServer.

2 Requirements

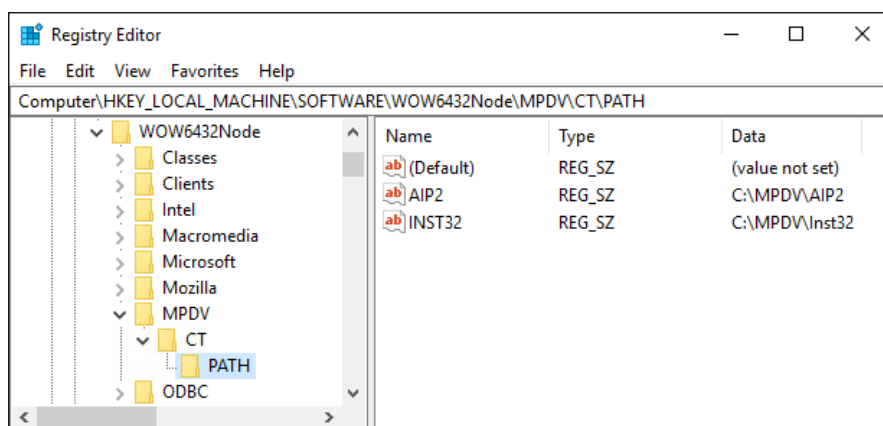
- For consistent datasets the HYDRA terminal software needs to run in 24/7 mode (24 hours per day, 7 days per week).
- The PC hardware and software must meet the recent recommendations of MPDV Mikrolab GmbH (see MPDV manual "HW_SW_GUIDE").
- Preinstalled Windows operating system approved by MPDV Mikrolab GmbH:
Windows 10 (32Bit or 64Bit).
(see MPDV manual "HW_SW_GUIDE").
- Latest Windows service pack approved by MPDV Mikrolab GmbH installed.
- A minimum of 500 MB capacity available on the disk drive (usual: C:\).
- Installed Internet Explorer version **6** or newer.
- Installed and functioning Windows Scripting (**WSH – Windows Script Host**).
- **TCP/IP** network configuration for the local network.
Network adapter must be up and running.
- The local Windows Firewall should be turned off.
- Installed **.NET Framework 3.5** (includes .NET 2.0 and 3.0).
- Installed **Visual C++ 2010 x86 Redistributable Package**.
Note: At least version 10.0.30319 of the Microsoft Visual C++ 2010 x86 Redistributable Package must be installed.
- Local Windows user (e.g.: "**hydadm**") with a secure password.
The password should be disclosed to MPDV.
The user must be a member of the local group Administrators.
Auto logon must be active for this user, see chapter "[4 Auto Logon for Windows](#)".

- Operating the Terminal without local administrator permissions will need some special configurations, see chapter “[7 Operating the Terminal without local administration permission](#)”.
- The Windows user who is supposed to run the AIP 8.2 Terminal software later will always need write permission for the chosen installation directories for INST32 and AIP 8.2.

Note:

After a successful installation as described in this manual those directories are stored in the Windows Registry:

[HKEY_LOCAL_MACHINE\SOFTWARE\Wow6432Node\MPDV\CT\PATH]



This installation instruction assumes that

- up to and including MES Weaver 3.1: HYDRA Nameserver
- from MES Weaver 4.0pe onwards: MIP Nameserver

is listening on network port 10000. You are tasked to check the actual port of your HYDRA installation. If the port of your HYDRA installation differs, you have to set a permanent environment variable in the AIP host's operating system:

HYPOR = ACTUAL_PORT_NUMBER

3 Pre Installation Configuration

3.1 Windows 10

- Create a local Windows user (e.g.: "hydadm") with a secure password.

The password should be disclosed to MPDV.

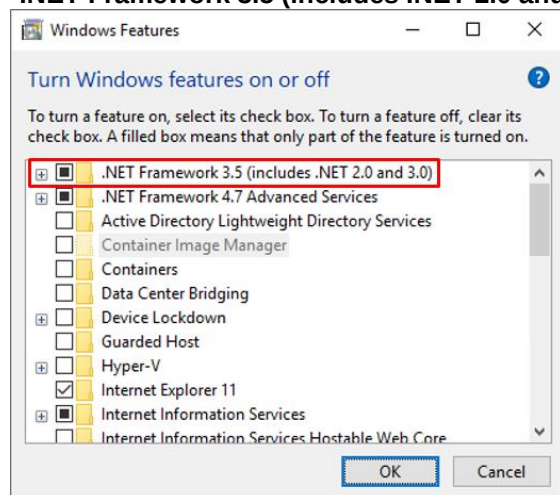
The user must be a member of the local group **Administrators**.

Auto logon must be activated for this user, see chapter "[4 Auto Logon for Windows](#)".

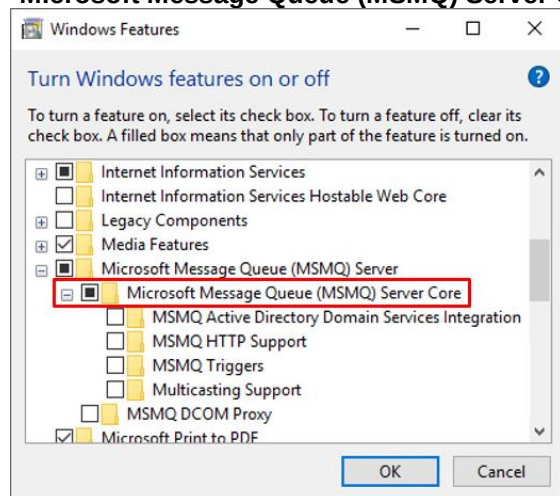
- Activate the following Windows Features:

Start-Button – Windows System – Control Panel – Programs – Programs and Features – Turn Windows features on or off

".NET Framework 3.5 (includes .NET 2.0 and 3.0)"



"Microsoft Message Queue (MSMQ) Server Core"



OK

- Install the Microsoft **Visual C++ 2010 x86 Redistributable Package**.

e.g. download from:

<https://www.microsoft.com/en-us/download/details.aspx?id=5555>

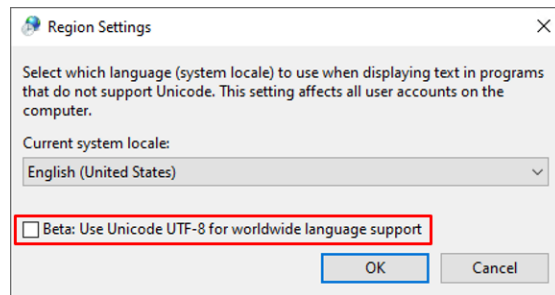
The installation package can also be found on the HYDRA server, e.g. in:

d:\mip1\inbetr\windows\msvc100\vc_redist_x86.exe

Note: At least version 10.0.30319 of the Microsoft Visual C++ 2010 x86 Redistributable Package must be installed.

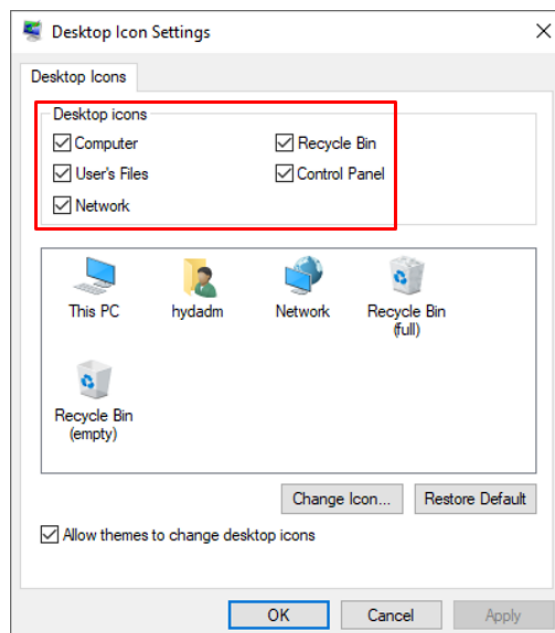
- **Start-Button – Windows System – Control Panel – Clock and Region – Region – Administrative – Change system locale...**

If there is the (Beta) checkbox “Use Unicode UTF-8 for worldwide language support” available make sure it is **not activated**:



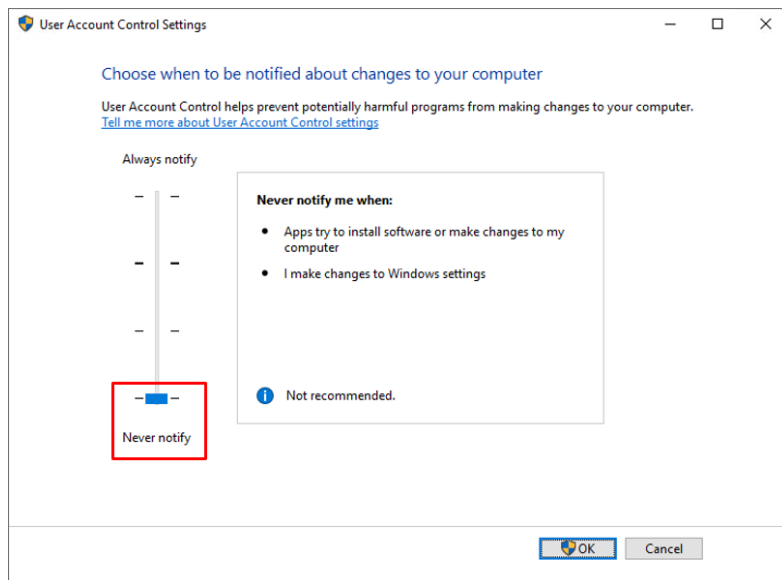
- **Start-Button – Settings – Themes – Desktop icon settings:**

Activate all Desktop icons:



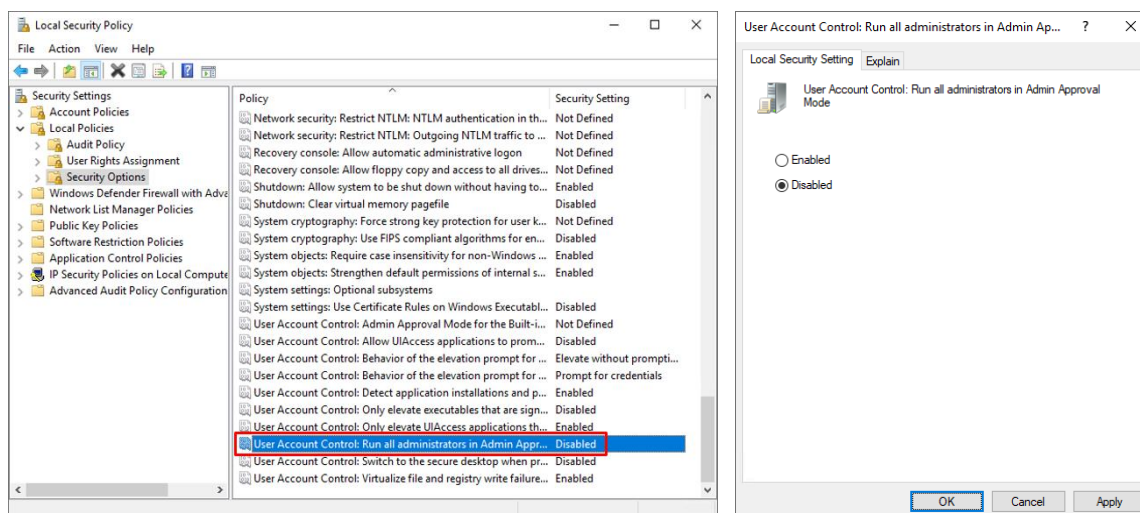
OK

- User Account Control Settings (UAC) must be **deactivated**:
Start-Button – Windows System – Control Panel – User Accounts – User Accounts – Change User Account Control settings



Never notify

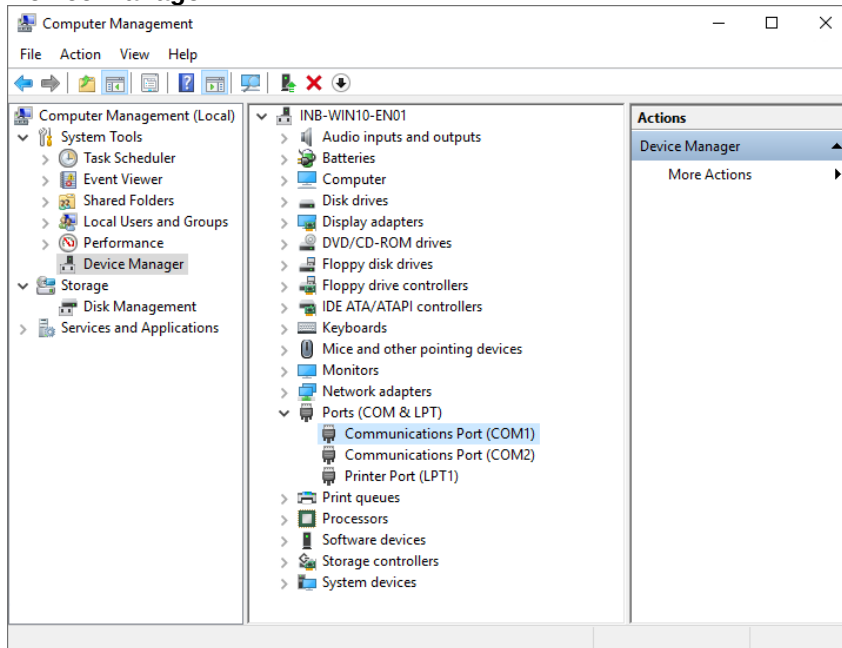
- In “Local Security Policy” the following policy must be **disabled**:
Start-Button – Windows System – Windows Administrative Tools – Local Security Policy – Local Policies – Security Options – User Account Control: Run all administrators in Admin Approval Mode



Disabled

- Check and adjust settings for all available COM ports and printer ports.

Start-Button – Windows System – Windows Administrative Tools – Computer Management – Device Manager

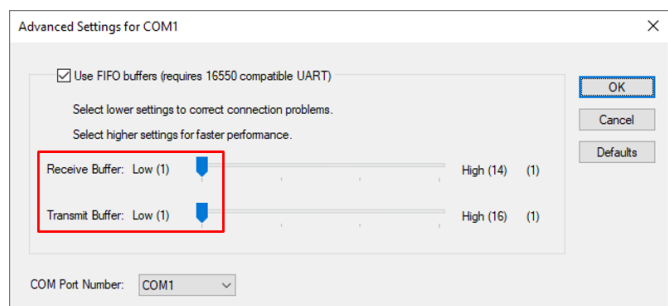
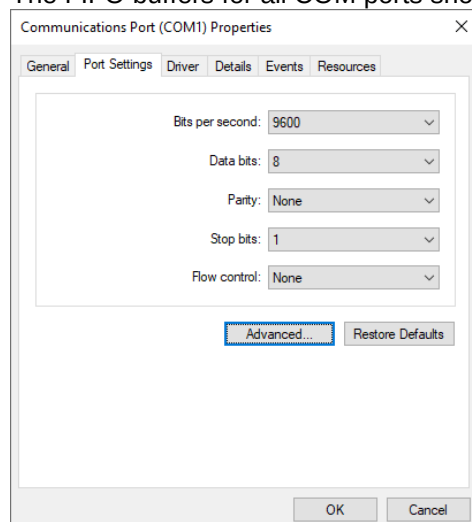


e.g.:

Communication Port (COM):

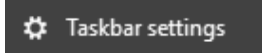
COM1	I/O-Address: 03F8	IRQ: 4
COM2	I/O-Address: 02F8	IRQ: 3
COM3	I/O-Address: 03E8	IRQ: 10
COM4	I/O-Address: 02E8	IRQ: 11

The FIFO buffers for all COM ports should be set to **low**:

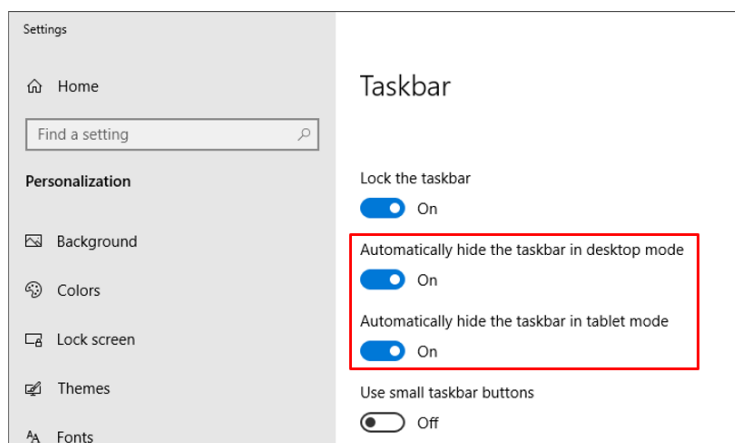


Should there be problems with serial connections the buffers might be set to higher levels again.

- Right click on Windows Taskbar and select "Taskbar settings":



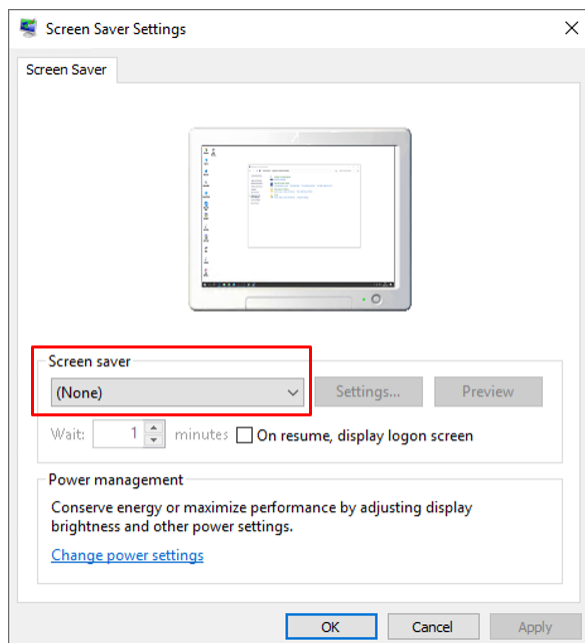
Activate settings for "Automatically hide the taskbar in desktop mode" and "Automatically hide the taskbar in tablet mode":



- Make sure there is no screen saver active.

Start-Button – Settings – Personalization – Lock Screen – Screen saver settings

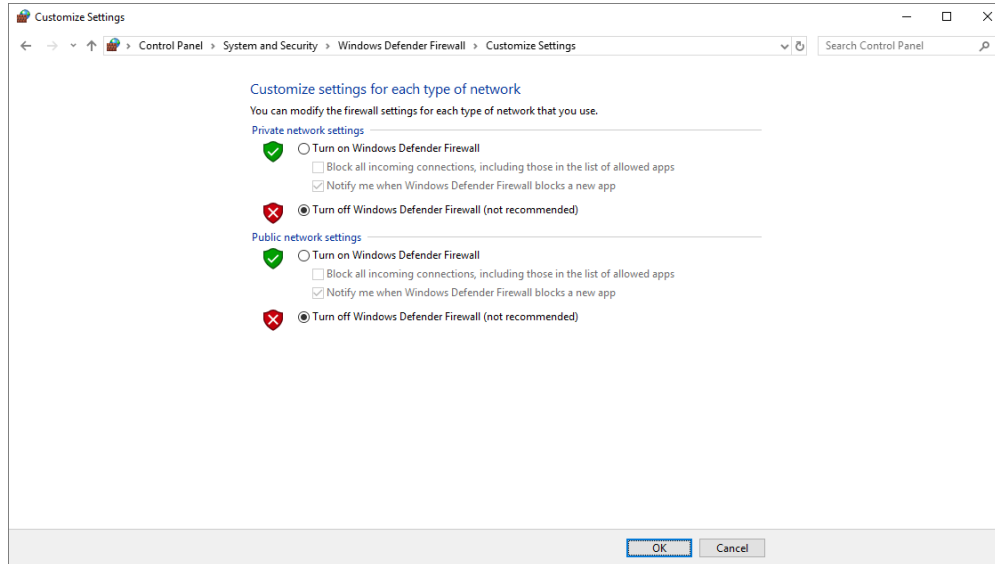
Screen saver: **(None)**



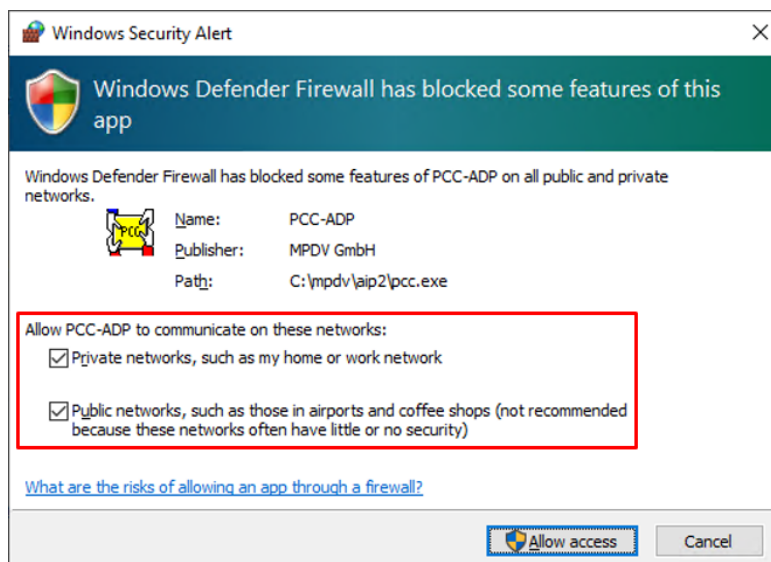
OK

- The local Windows Firewall should be turned off:

Start-Button – Windows System – Control Panel – System and Security – Windows Defender Firewall – Turn Windows Defender Firewall on or off



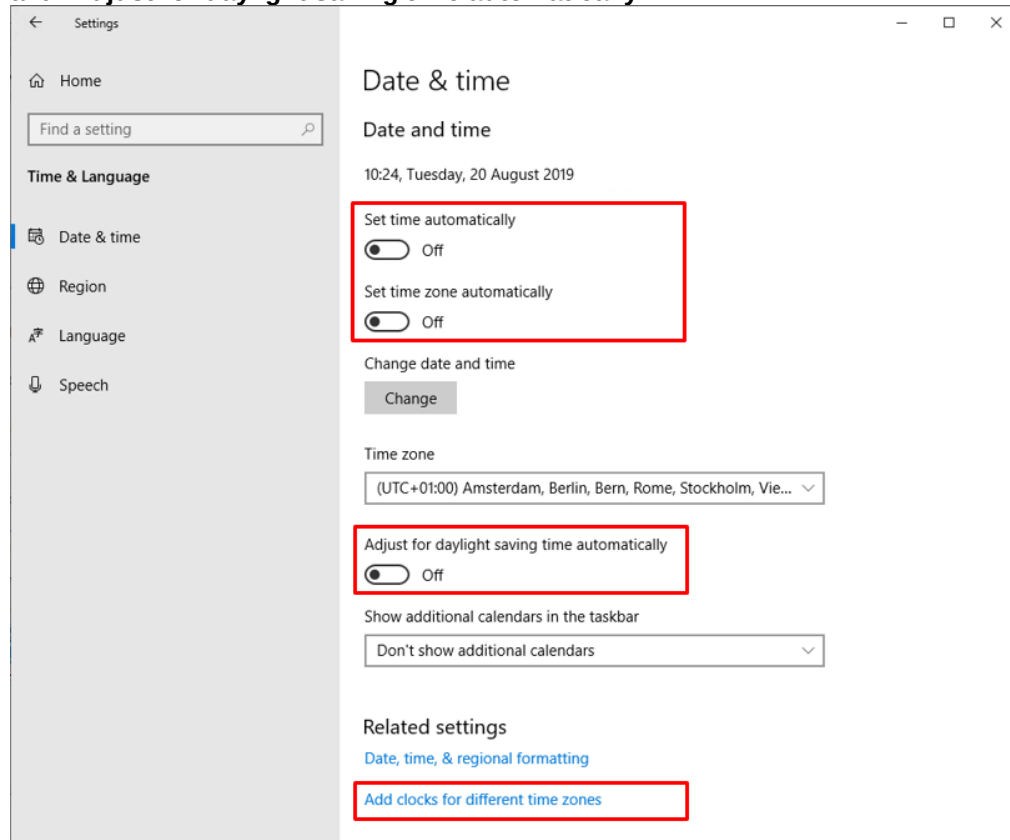
If the local firewall stays active you might need to manually allow communication access for several programs like pcc.exe and ctai.p.exe when starting the AIP terminal software for the first time.



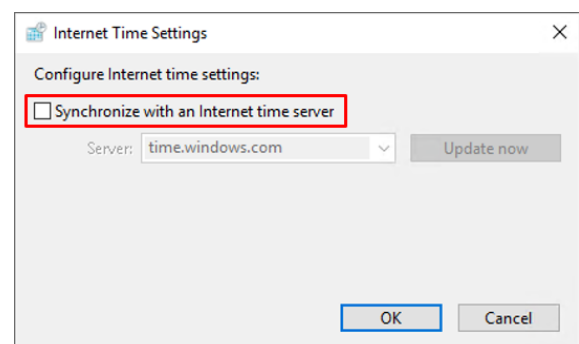
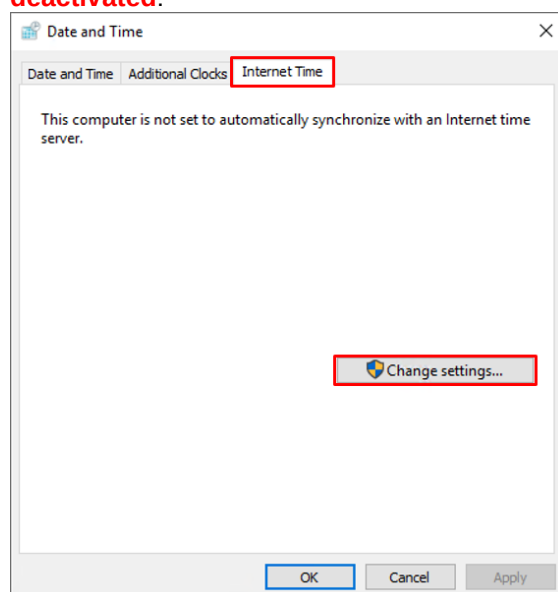
- Change date and time settings:

Start-Button – Settings – Time & Language

Deactivate all automatic time settings, like “Set time automatically”, “Set time zone automatically” and “Adjust for daylight saving time automatically”:

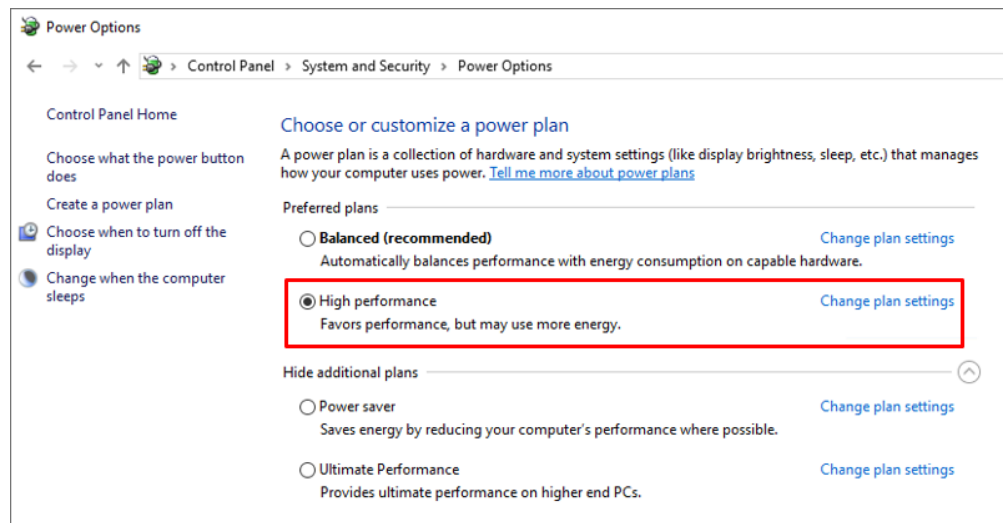


In “Add clocks for different time zones” make sure that synchronization with a time server is **deactivated**:



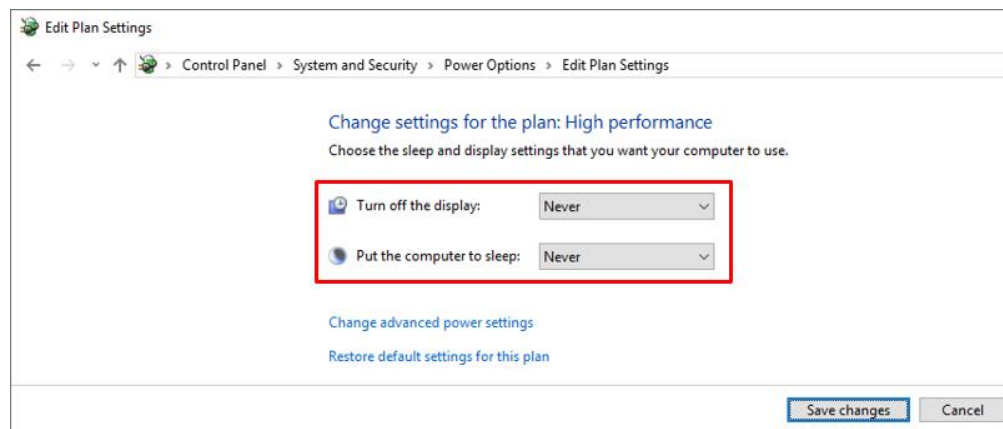
- Activate power plan **“High performance”**:

Start-Button – Windows System – Control Panel – System an Security – Power Options

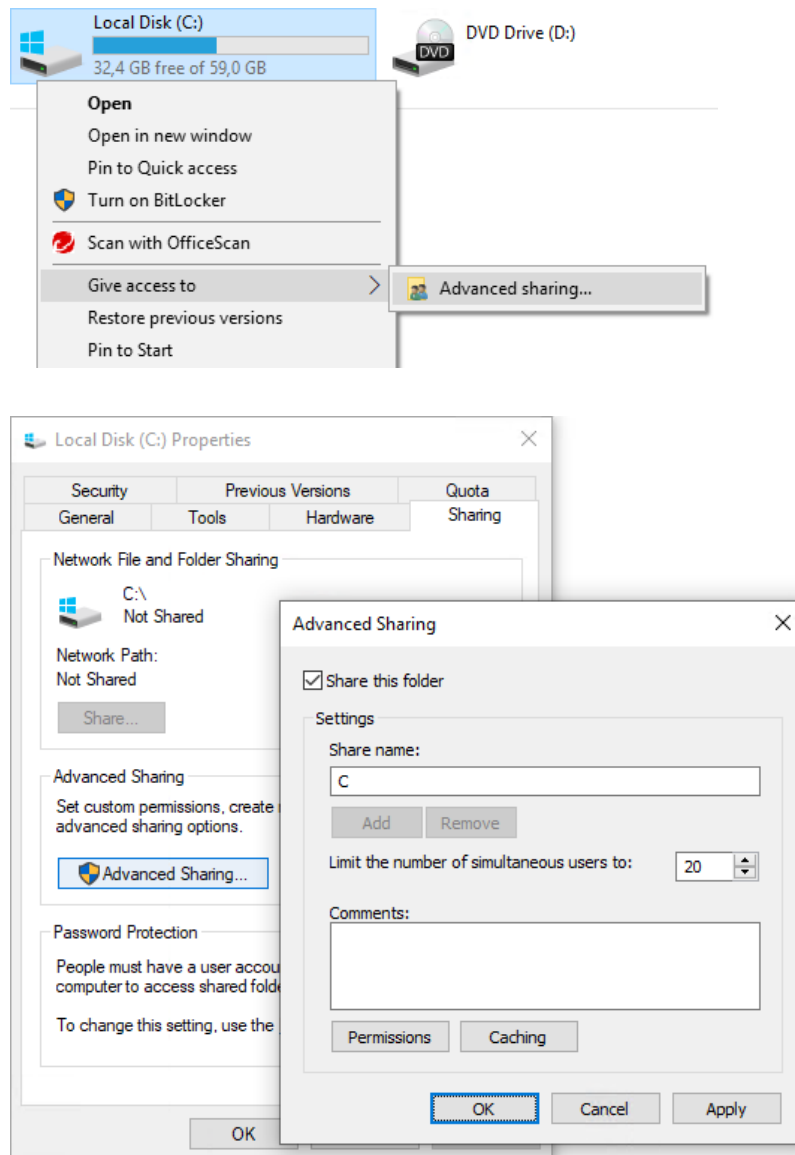


In "Change plan settings":

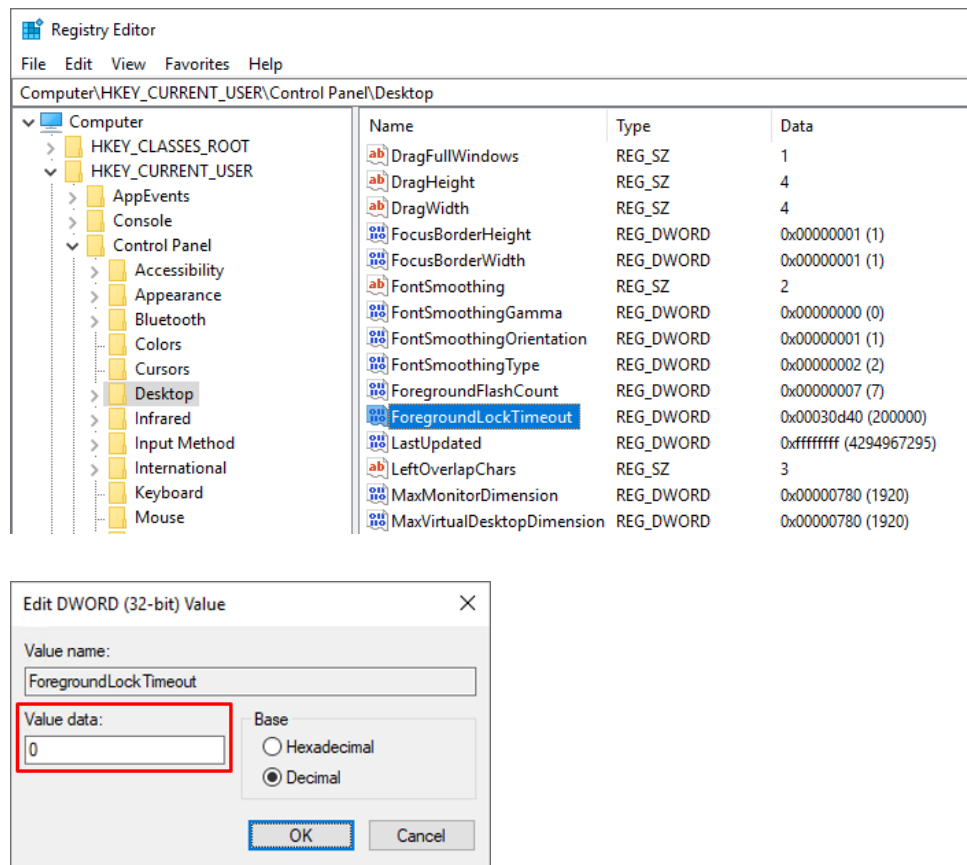
Set **Never** for "Turn off the display" and "Put the computer to sleep":



- For support purposes it is recommended to activate the network sharing for the disk drive where the HYDRA Terminal software is installed (usual: C:\).



- In the Windows registry the value for "ForegroundLockTimeout" within the key "HKEY_CURRENT_USER\Control Panel\Desktop" must be set to "0":

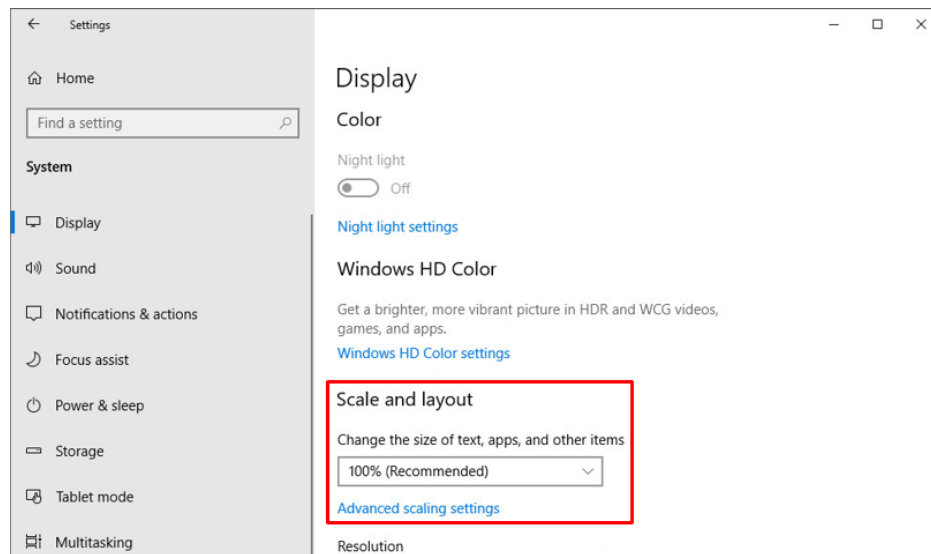


This value specifies the time (in milliseconds) during which the system keeps applications from moving to the foreground.

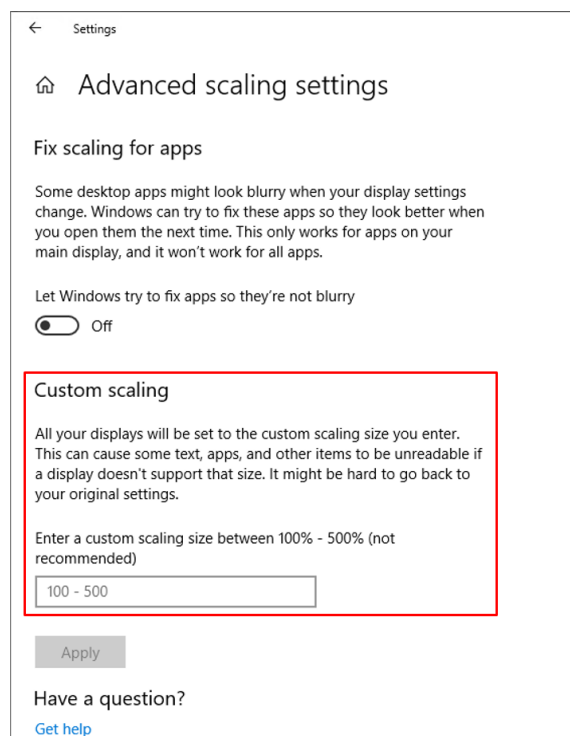
With that setting all terminal dialogues waiting for user input are moved to the foreground immediately.

- The size of text, apps and other items must be set to the recommended default value **100%**:

Start-Button – Settings – System – Display



In “Advanced scaling settings” make sure that no custom scaling is active:



When using other scaling sizes than 100% you will encounter overlapping text in the terminal dialogues.

4 Auto Logon for Windows

For certain functionality of the HYDRA terminal software, e.g. like the automatic restart (see chapter "[6.5 Automatic Restart](#)") or remotely controlled restarts from a MOC client, it is necessary to have the auto logon functionality for Windows enabled.

Alternatively to the methods described below the tool "Autologon for Windows v3.10" from Microsoft can be used to activate the auto logon mechanism of Windows:

<http://technet.microsoft.com/en-us/sysinternals/bb963905>

At least version v3.01 must be used.

The tool runs on:

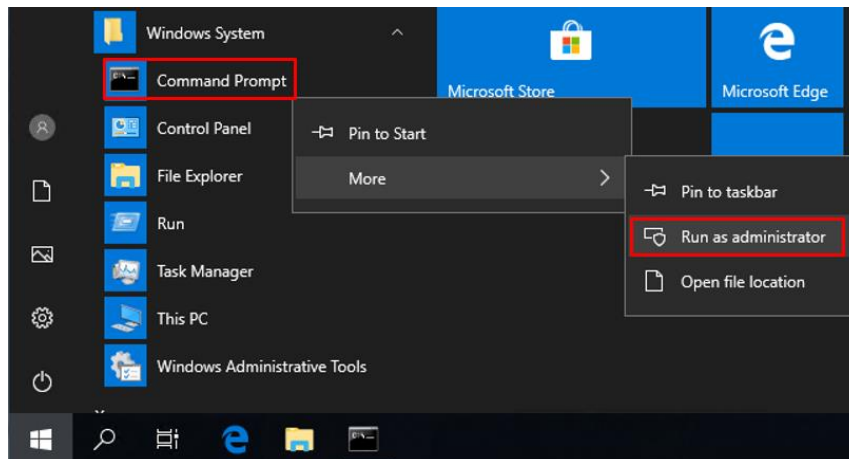
Client: Windows XP and higher

Server: Windows Server 2003 and higher

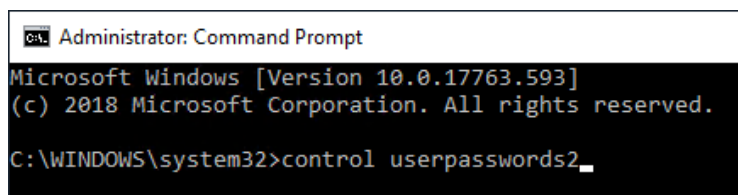
4.1 Windows 10

- **Start-Button – Windows System**

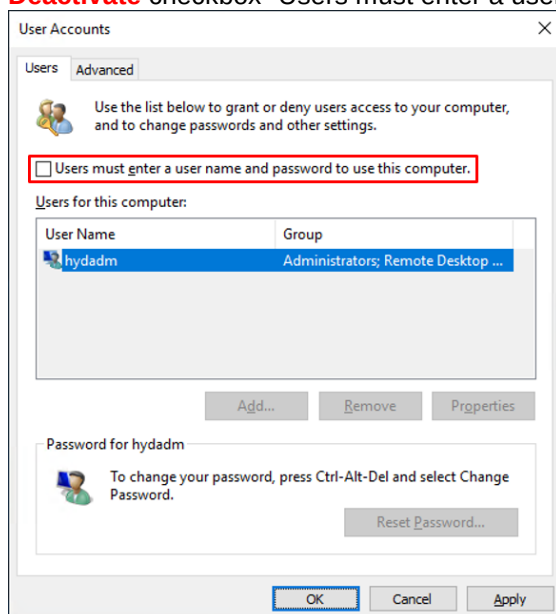
Run Windows **Command Prompt as Administrator**:



- Execute **"control userpasswords2"**

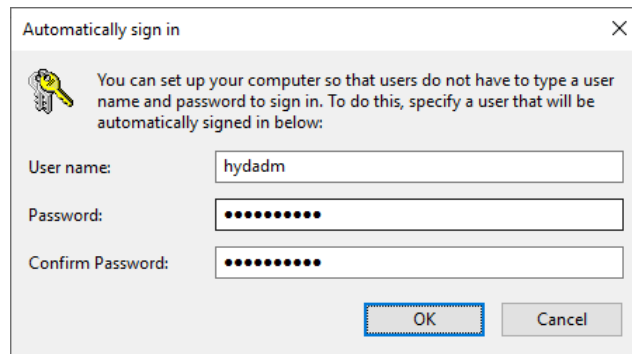


Deactivate checkbox "Users must enter a user name and password to use the computer"



OK

- Set "User name" and "Password" for automatic sign in (e.g. user **hydadm** and his secure password).



Automatically sign in

You can set up your computer so that users do not have to type a user name and password to sign in. To do this, specify a user that will be automatically signed in below:

User name:

Password:

Confirm Password:

OK Cancel

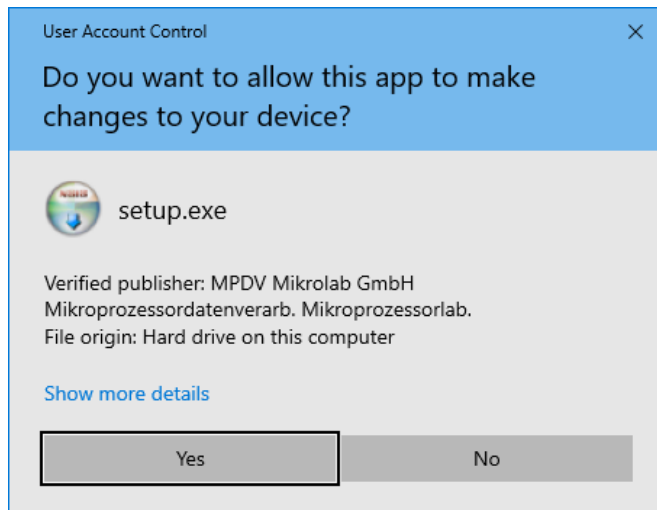
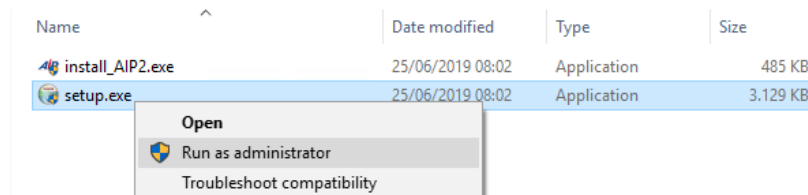
OK

- **Restart your computer now.**
Check that Windows is starting without asking for a user name and password.

5 Installation of AIP 8.2

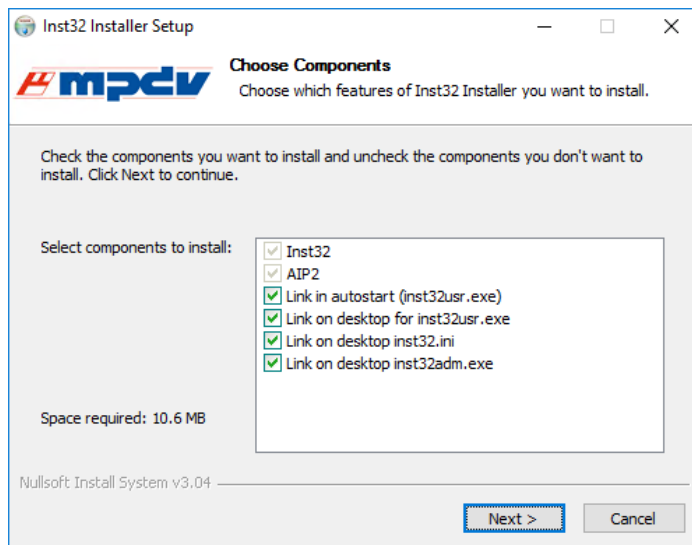
5.1 Windows 10

- If Auto Logon for Windows might not be enabled yet (see chapter "[4 Auto Logon for Windows](#)"), login as local Windows user (e.g.: **hydadm**) which must be a member of the local group **Administrators**.
- Copy the directory "**inst_terminal_mw4**" with all its contents from the HYDRA server to a temporary directory on your Terminal PC.
"inst_terminal_mw4" can be found on the HYDRA server in the directory %HYDRADIR%\admttools (e.g. d:\mip1\admttools\inst_terminal_mw4).
- In the local copy of directory "inst_terminal_mw4" start the file **setup.exe** using the option "**Run as administrator**" and then proceed with the installation as described below:



Yes

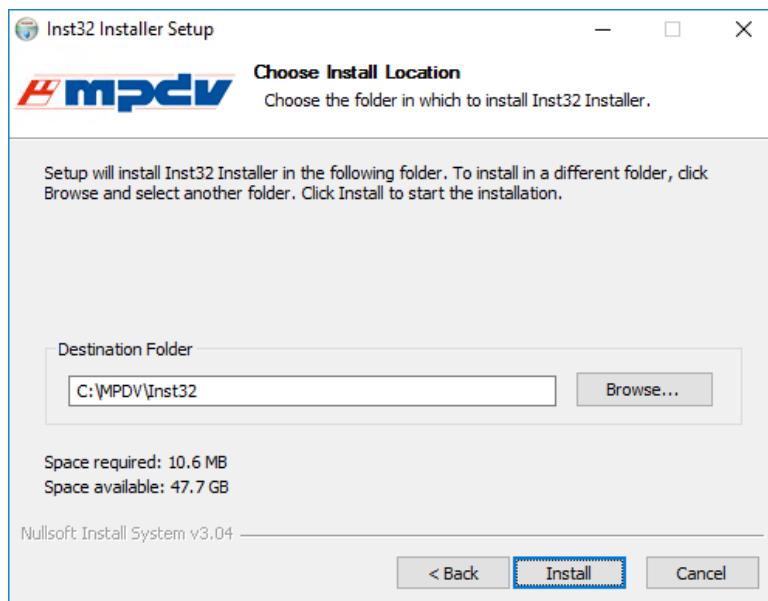
Select components for the Inst32 Installer (**select all as default**):



Deselect components only if you are absolutely aware of the consequences.

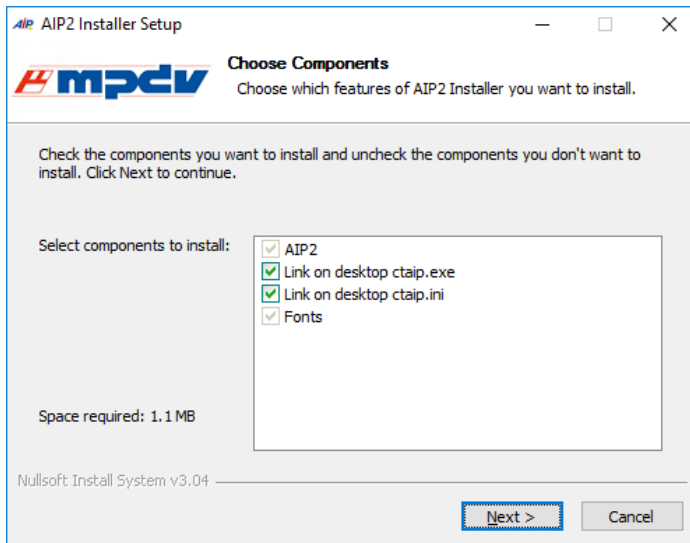
Next

Choose destination folder for the Inst32 installer (Default: C:\MPDV\Inst32):



Install

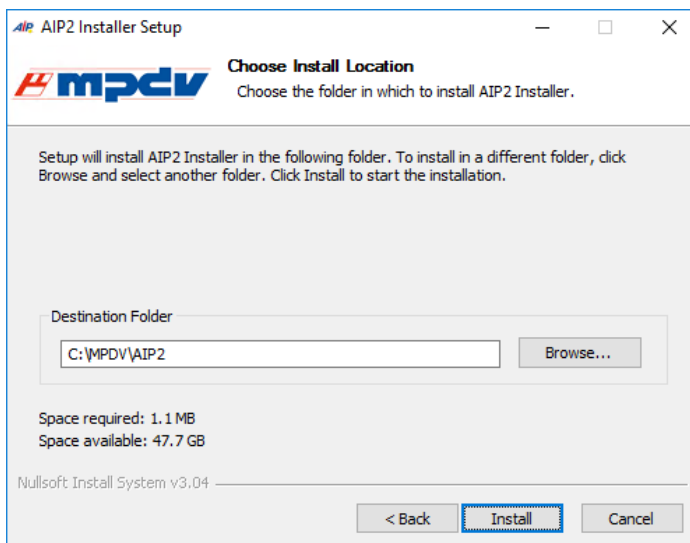
Select components for the AIP2 Installer (**select all as default**):



Deselect components only if you are absolutely aware of the consequences.

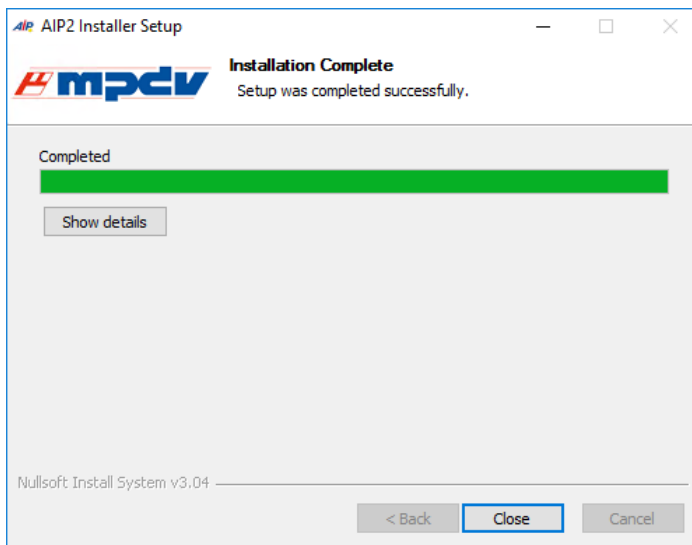
Next

Choose destination folder for the AIP2 installer (Default: C:\MPDV\AIP2):

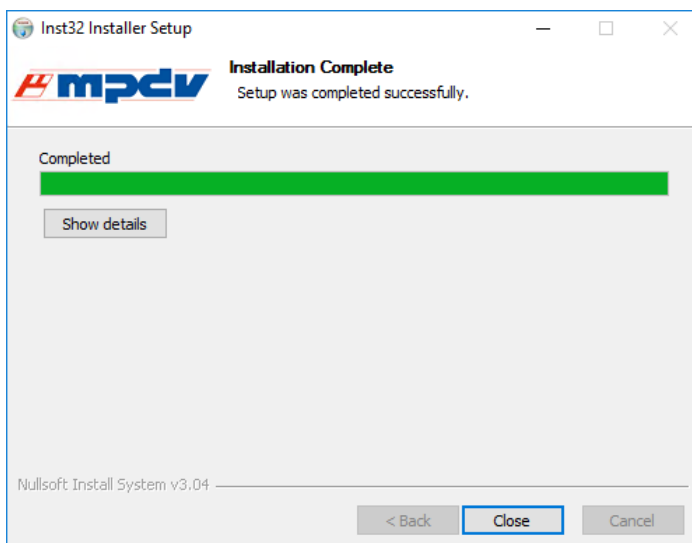


Install

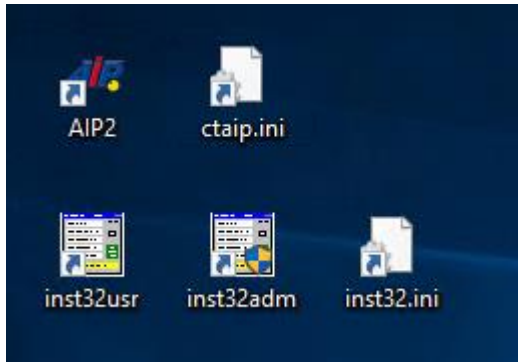
Close the AIP2 Installer

**Close**

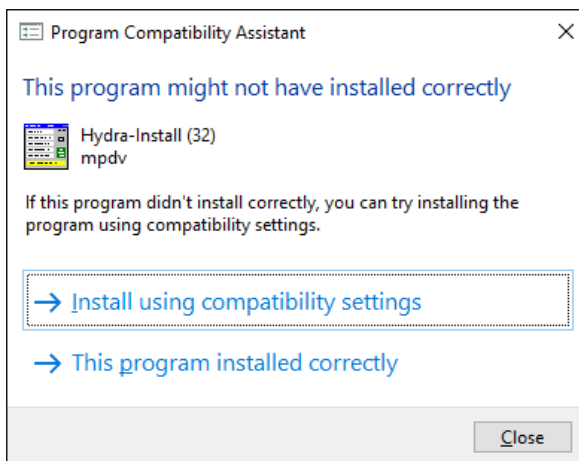
Close the Inst32 Installer

**Close**

- Now you should find the following shortcuts on the Windows Desktop:



- When you start and close Inst32adm.exe or Inst32usr.exe for the first time you might see a message like this:



Select **"This program installed correctly"**

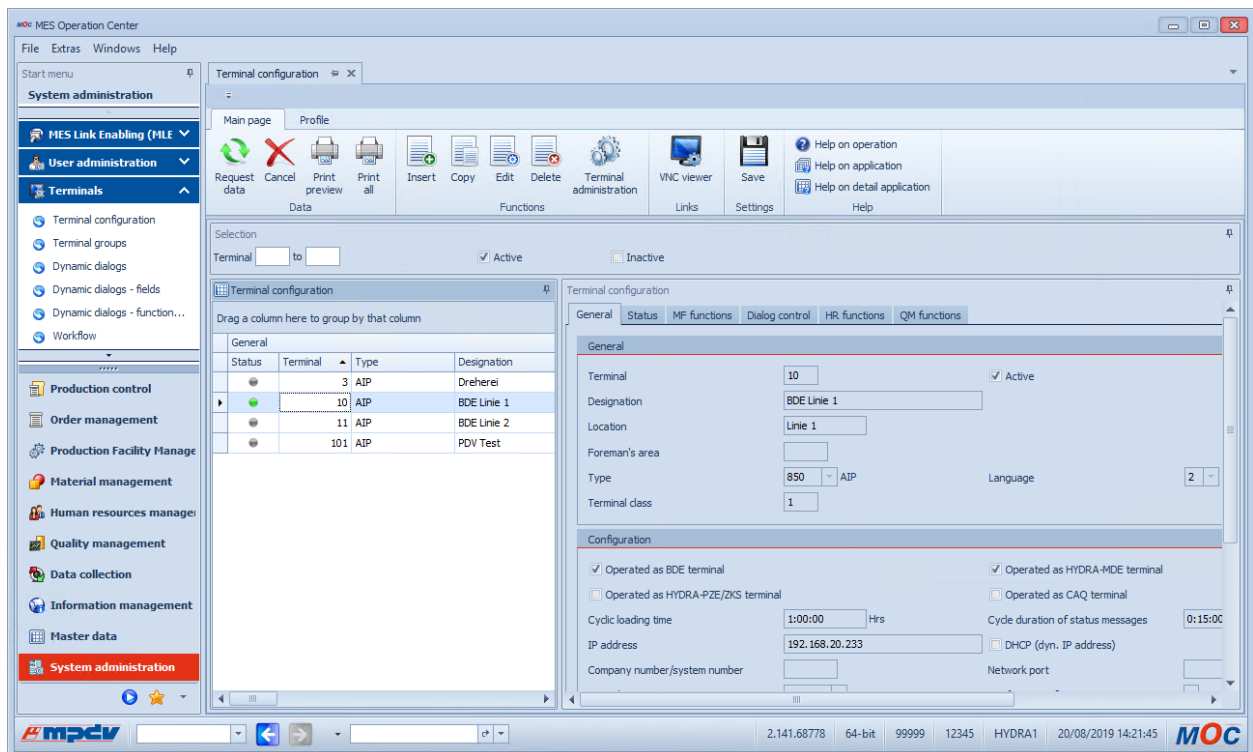
- Before the application AIP 8.2 can be started you need to proceed with chapter "[6 Post Installation Configuration](#)" first.

6 Post Installation Configuration

6.1 Terminal Configuration Set

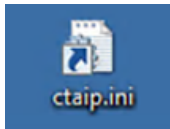
Before you can start your terminal client you have to create a terminal configuration set first by using the HYDRA MOC client.

System administration → Terminals → Terminal configuration



6.2 AIP 8.2 Configuration

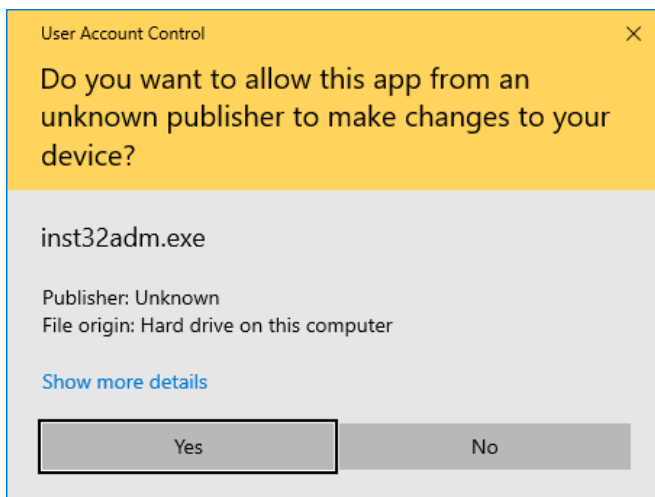
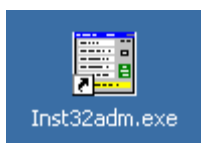
- Open the file **ctaip.ini** by double clicking on the Icon on the Windows Desktop:



- In section [system] enter the IP-address or the hostname of your HYDRA server and then save the file:

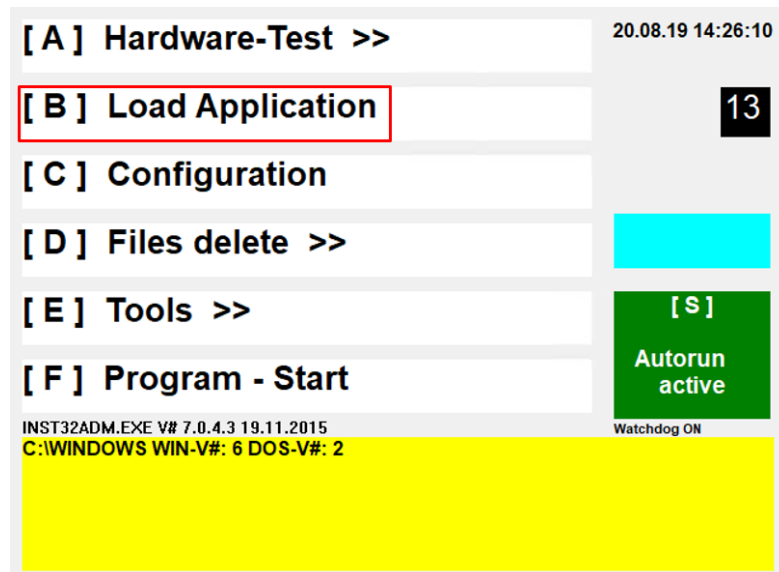
```
[system]
usr=300
hostname=10.10.60.86
loadfile=ctnet\win\aip2.txt
```

- Start **Inst32adm.exe** by double clicking on the Icon on the Windows Desktop:

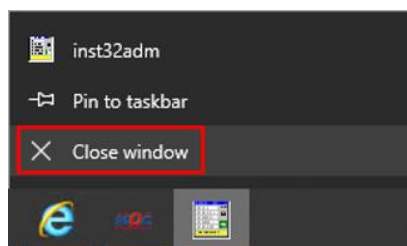


Yes

- Choose the menu **[B] Load Application**



- After successfully downloading all the program files close the program Inst32adm.exe:
Do not start the AIP2 terminal software yet!



- Delete the file **ctaip.ini** in the installation directory of the terminal (e.g.: C:\MPDV\aip2).
- Inside the installation directory of the terminal (e.g.: C:\MPDV\aip2) rename the file **ctaip.bsp** to **ctaip.ini**.
- Open the (new) file **ctaip.ini** by double clicking on the Icon on the Windows Desktop and enter correct values in section [system], e.g.:

```
usr=10
```

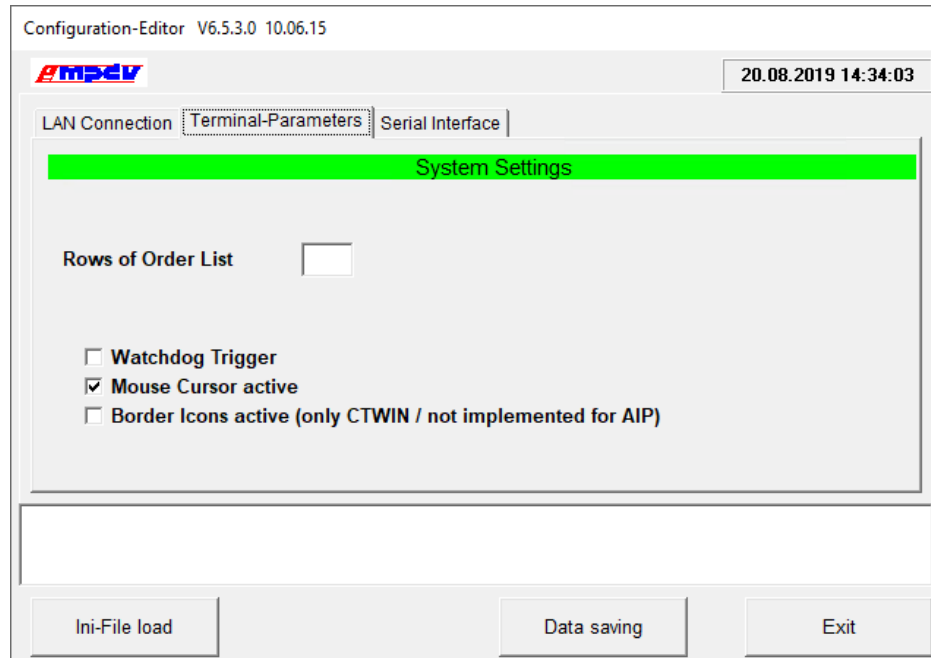
(Terminal ID, see chapter "[6.1 Terminal Configuration Set](#)")

```
hostname=192.168.20.167
```

(IP-address or host name of the HYDRA Server)

- Start **Inst32adm.exe** by double clicking on the Icon on the Desktop and choose the menu **[C] Configuration:**

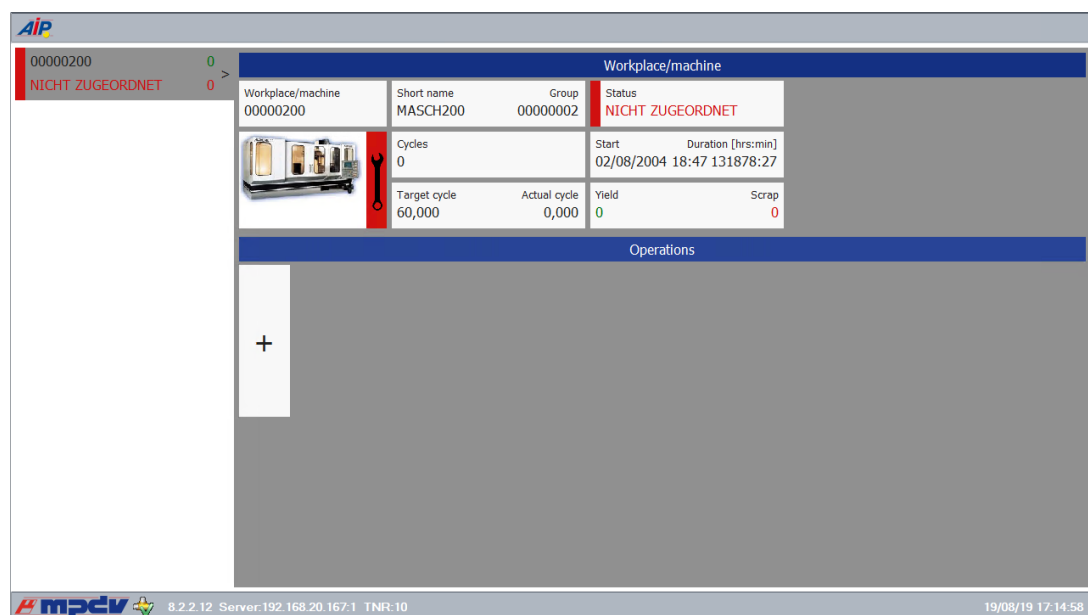
For use on a standard PC the option "**Mouse Cursor active**" must be activated:



Data saving

Exit

- The basic installation of the terminal software is finished now.
The terminal can be started by using menu **[F] Program – Start**



6.3 Automatic Clock Adjustment

The HYDRA terminal software AIP 8.2 is synchronizing the local clock of the terminal PC with the time of the HYDRA server.

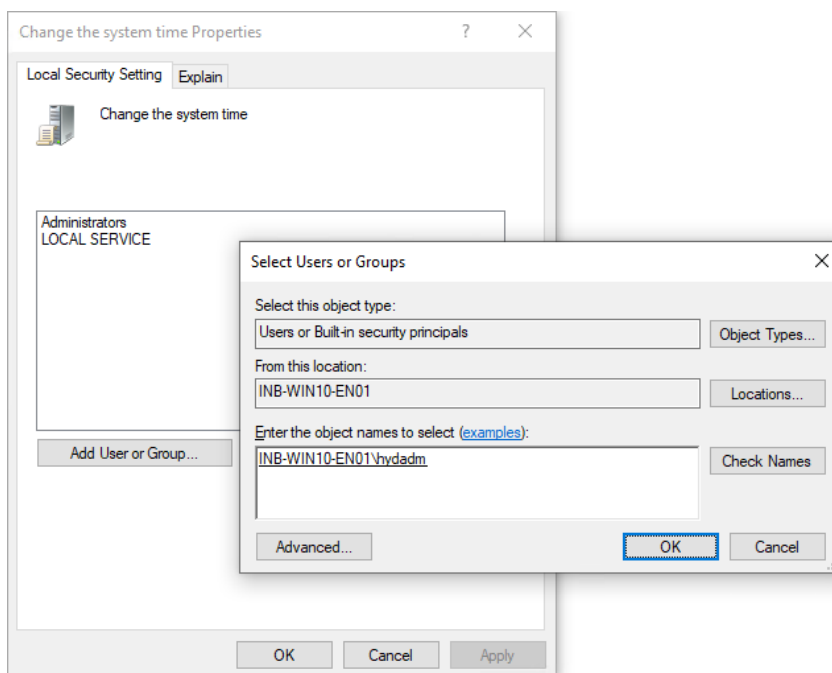
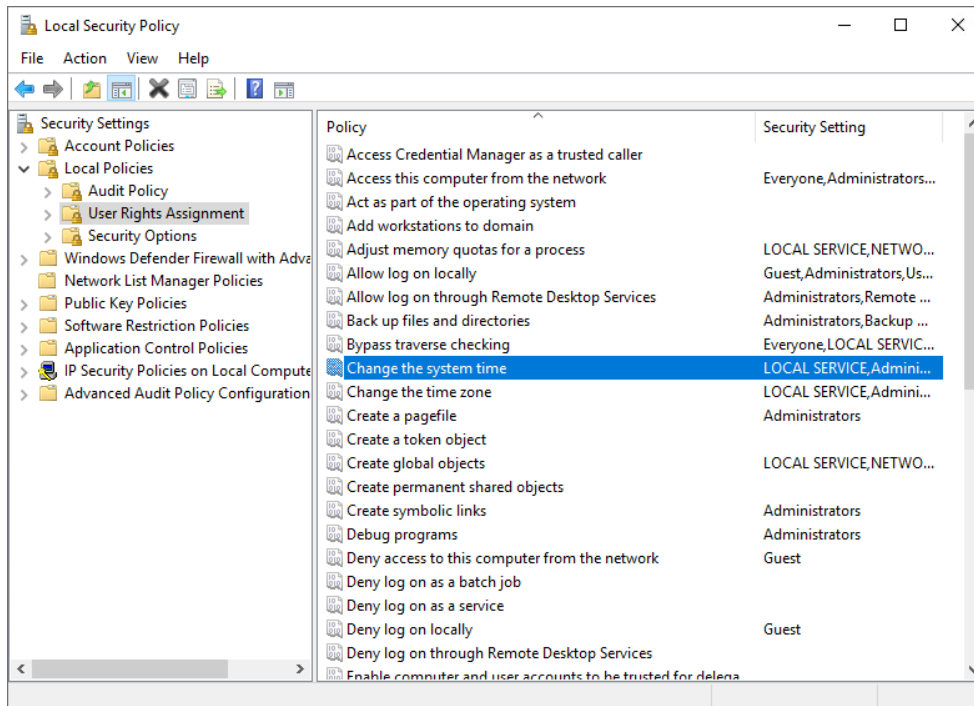
The Windows user running the terminal software must be able to set the local system time.

Should the local Windows user account who is running the AIP 8.2 software do not have the required permissions you have to make sure that the system time of the HYDRA terminal PCs and the HYDRA server are synchronized by other means.

6.3.1 Windows 10

The permission to set the system time can be assigned to Windows users:

Start-Button – Windows System – Windows Administrative Tools – Local Security Policy – Local Policies – User Rights Assignment – Change the system time



Use "Add User or Group..." to add the user account

OK

6.3.2 Deactivating Automatic Clock Adjustment

Should the local Windows user account who is running the AIP 8.2 software do not have the required permissions to set the system time for any reasons or should the local clock be synchronized and set by other means, the automatic clock adjustment of the HYDRA terminal software must be deactivated.

Therefor you have to add he following settings to section **[system]** in the configuration file **CTAIP.INI**:

```
[system]
```

```
NoLocalWatchUpdate=on
```

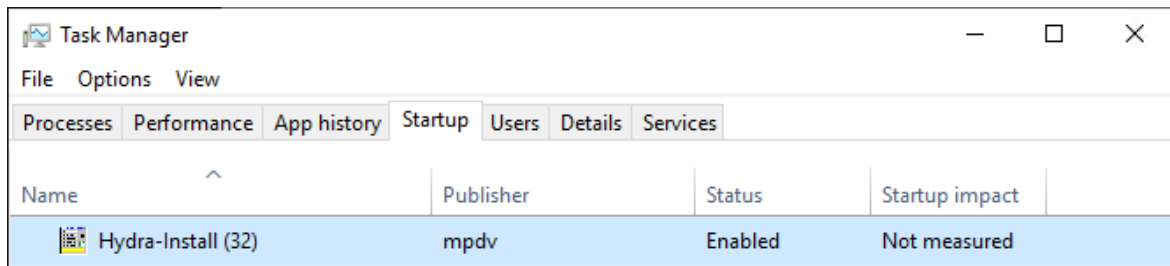
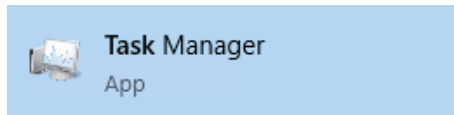
```
NoLocalWatchWarning=ON
```

When this configuration is used the system administrator has to make sure that the system time of the HYDRA terminal PCs and the HYDRA server are synchronized by other means.

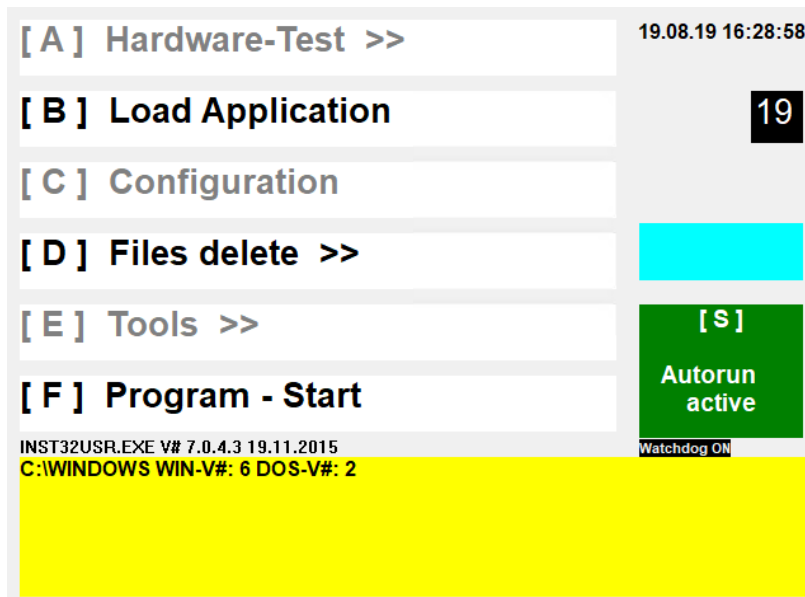
6.4 Auto Start AIP 8.2

To ensure an automatic startup of the AIP terminal software the installer added inst32usr.exe to the Windows Task Manager:

Start-Button – Task Manager



Whenever a user is logging into Windows, **inst32usr.exe** will start automatically:



Subsequently AIP 8.2 will then be started by inst32usr.exe.

6.5 Automatic Restart

There is a build-in 32bit timer in Windows which increases every one msec.

Every 49,7 days that counter will create an overflow which could cause strange behavior of the AIP 8.2 terminal software like hang-ups.

Therefore the HYDRA terminal software will reboot the terminal PC automatically on the first coming Sunday at 1:30am after a 30 day period.

If it might be necessary to deactivate the automatic restart of the HYDRA terminal add the following settings to section **[system]** in the configuration file **CTAIP.INI**:

```
[system]
BootTermin=Off
```

When this configuration is used the system administrator has to make sure that the terminal PC gets restarted on a regular basis before that 49,7 days elapses.

6.6 Gateway Communication

By default the gateway communication option for the HYDRA terminal is active and requires access to port 9002.

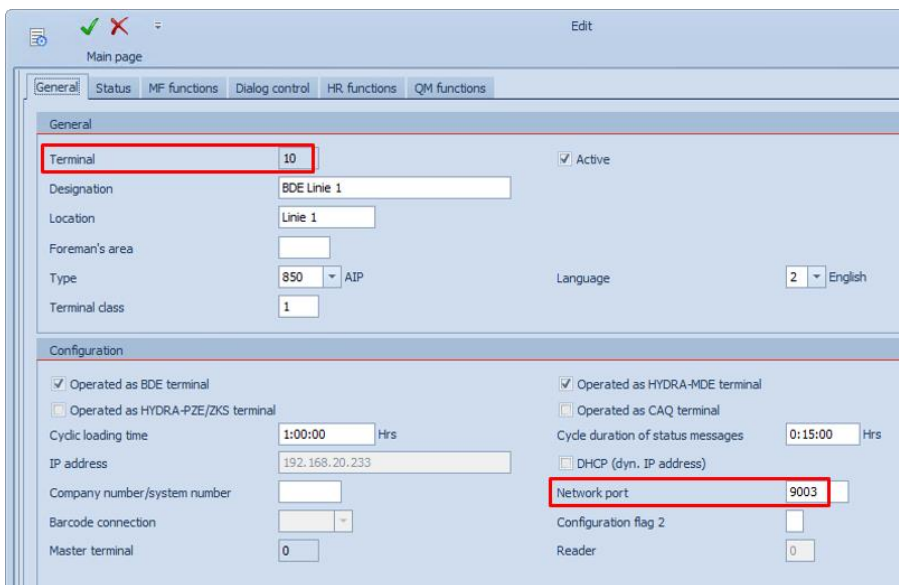
If the gateway communication should use a different port (e.g. 9003) it is necessary to assign a unique port to the terminal by adding the following settings to section **[GateWay-Communication]** in the configuration file **CTAIP.INI**:

[GateWay-Communication]

Active=true

Port=9003

Make sure that the specified port is defined in the corresponding terminal configuration set of your AIP2 terminal (see chapter [6.1 Terminal Configuration Set](#)):



The screenshot shows the AIP2 terminal configuration window. The 'General' tab is active, showing fields for Terminal (10), Designation (BDE Linie 1), Location (Linie 1), Foreman's area, Type (850 AIP), Language (2 English), and Terminal class (1). The 'Configuration' tab is also visible, showing options for operating as BDE, HYDRA-PZE/ZKS, or HYDRA-MDE terminal. The 'Network port' field is set to 9003 and is highlighted with a red box.

To deactivate the gateway communication option for the HYDRA terminal add the following settings to section **[GateWay-Communication]** in the configuration file **CTAIP.INI** (create that section if it does not exist):

[GateWay-Communication]

Active=false

6.7 Barcode or ID-Code Readers

Depending on which COM port the assigned reader hardware is connected add the appropriate settings to section **[comports]** in the configuration file **CTAIP.INI**.

e.g.:

```
[comports]
com5=BAR
com6=LEGIC
```

By default AIP2 is scanning for up to 30 COM ports.

If there might be more than 30 COM ports available on your terminal PC you have to add the COMCOUNT setting to section **[comports]** in the configuration file **CTAIP.INI**.

The value for COMCOUNT depends on the available amount of COM ports at the terminal PC.

e.g.:

```
[comports]
COMCOUNT=60
com45=BAR
com46=LEGIC
```

6.8 Operating without Touch Screen Monitor

AIP2 is especially designed to run on industrial PCs with a touch screen monitor and without a standard keyboard and mouse connected.

Therefore AIP2 is hiding the Windows mouse cursor by default.

If the terminal PC does not provide a touch screen monitor you need to tell AIP2 not to hide the Windows mouse cursor.

That will improve the usability on such an environment.

Add the following settings to section **[system]** in the configuration file **CTAIP.INI**:

```
[system]
showcursor=ON
```

For use on a standard PC that option must be activated.

6.9 Restricting AIP2 Window Size

By default AIP2 is using the available screen resolution provided by the Windows operating system so its window size will use up the whole screen.

If it should be necessary to restrict the AIP2 window size add the following settings to section **[system]** in the configuration file **CTAIP.INI**:

e.g.:

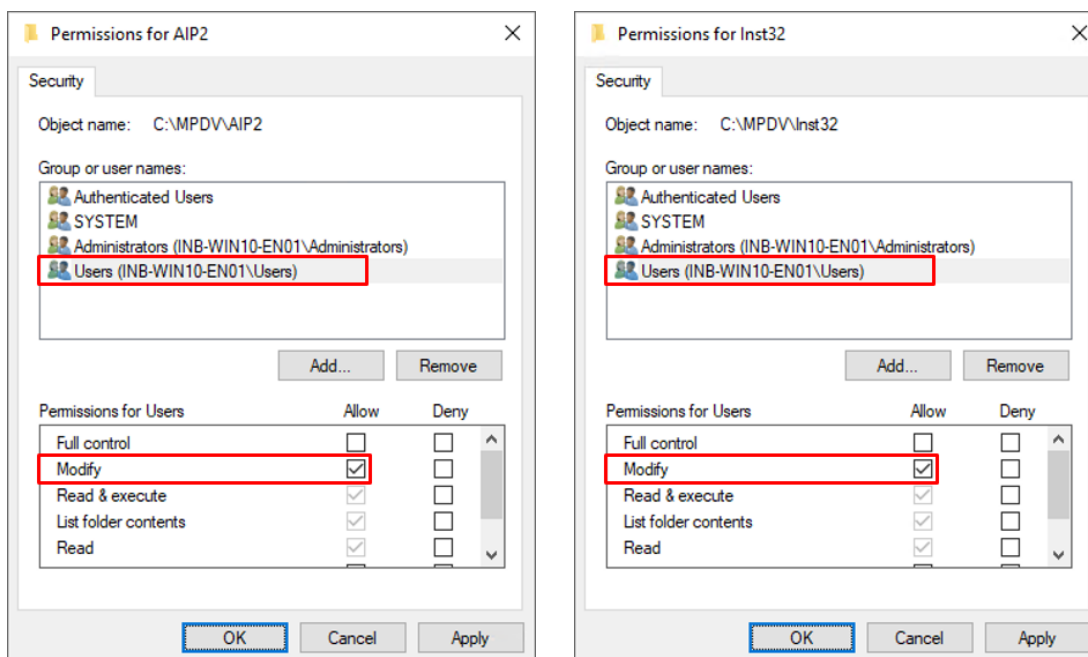
```
[system]
VirtScreenSize=1280
```

7 Operating the Terminal without local administrator permission

Basically it is possible to run the AIP Terminals without local administrator permissions. Depending on the used HYDRA modules and drivers, special actions might be necessary.

Additionally the measures about automatic clock adjustment described in [chapter 6.3](#) must be followed.

Grant permission "**Modify**" to the Windows group "**Users**" for the AIP and Inst32 installation directories (e.g.: "**C:\MPDV\AIP2**" and "**C:\MPDV\Inst32**") and for all their contents:



7.1 Information on PDV

Operating the PDV is possible without local administrator permission, just the component for the Online-Visualization has to be registered with local administrator permission.

Please open a command prompt (as local administrator) and run the following command:

32 bit Windows

```
regsvr32 c:\MPDV\AIP2\mpdvvisualisationclient.ocx
```

64 bit Windows

```
%systemroot%\SysWoW64\regsvr32 c:\MPDV\AIP2\mpdvvisualisationclient.ocx
```

8 Activation of Label Printing

A valid license for label printing needs to be installed on the HYDRA server.

Please note: The inst32 program loads the program package llprinter2.zip from the following HYDRA server directory: .\ctnet\win\shared\llprinter2.zip (e.g.: d:\mip1\ctnet\win\shared\llprinter2.zip).

The AIP 8.2 client unpacks the file llprinter2.zip during the next startup.

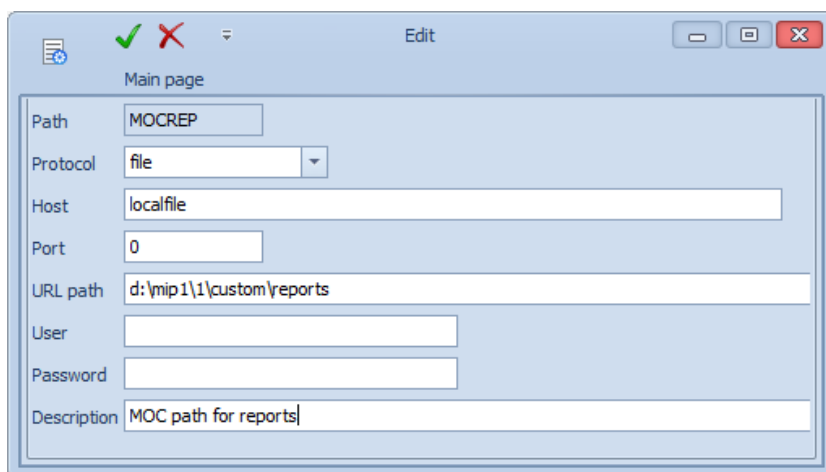
MOC - Path

For label maintenance, the HYDRA path MOCREP is required.

It is mandatory that the path always refers to <MDT>/custom/reports (e.g.: d:\mip1\1\custom\reports).

It is not allowed to change this.

Example configuration:

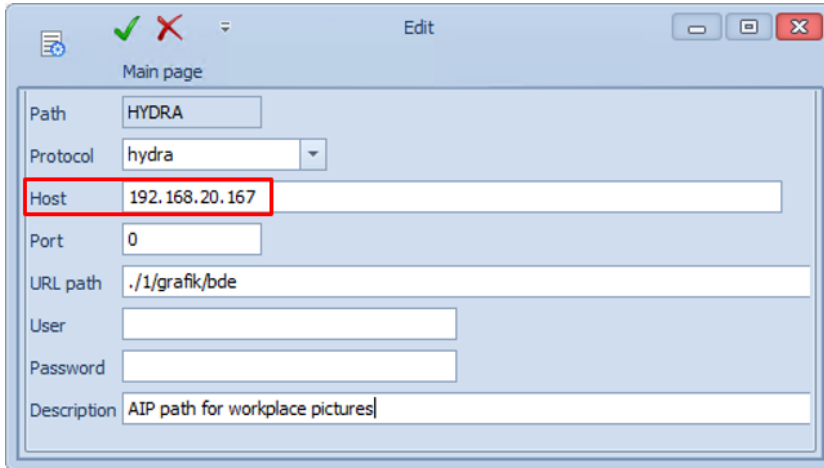


The screenshot shows a configuration window titled 'Main page' with a toolbar containing a green checkmark, a red X, and an 'Edit' button. The window contains the following fields:

Field	Value
Path	MOCREP
Protocol	file
Host	localfile
Port	0
URL path	d:\mip1\1\custom\reports
User	
Password	
Description	MOC path for reports

9 Machine pictures

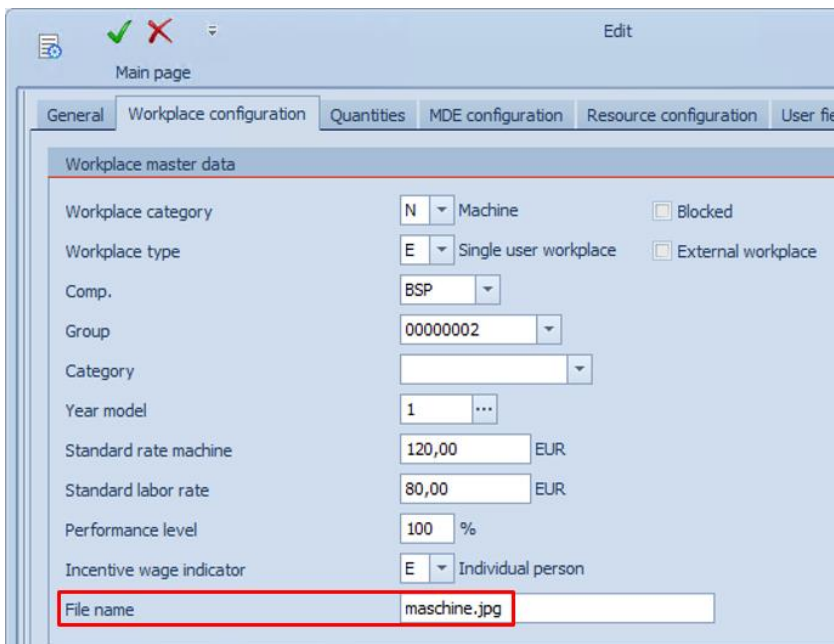
To display machine or workplace pictures in the AIP 8.2 terminal software the HYDRA path configuration for a path named “HYDRA” is required, e.g.:



Make sure you have entered the correct host name or the IP address of your HYDRA server.

According to this example all the picture files for your machines or workplaces (e.g. maschine.jpg) need to be stored inside the directory “./1/grafik/bde” on your HYDRA server, e.g.: d:\mip1\1\grafik\bde.

In the workplace configuration you need to enter the appropriate file name of the intended picture file, e.g. maschine.jpg:



Your machine or workplace picture will then be displayed in the AIP terminal software, e.g.:

AIP

00000200

0

>

NICHT ZUGEORDNET

0

Workplace/machine

Workplace/machine 00000200	Short name MASCH200	Group 00000002	Status NICHT ZUGEORDNET
	Cycles 0	Start Duration [hrs:min] 02/08/2004 18:47 131878:27	
	Target cycle 60,000	Actual cycle 0,000	Yield Scrap 0 0

Operations

InstallationGuide_MW40_AIP2.docx

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